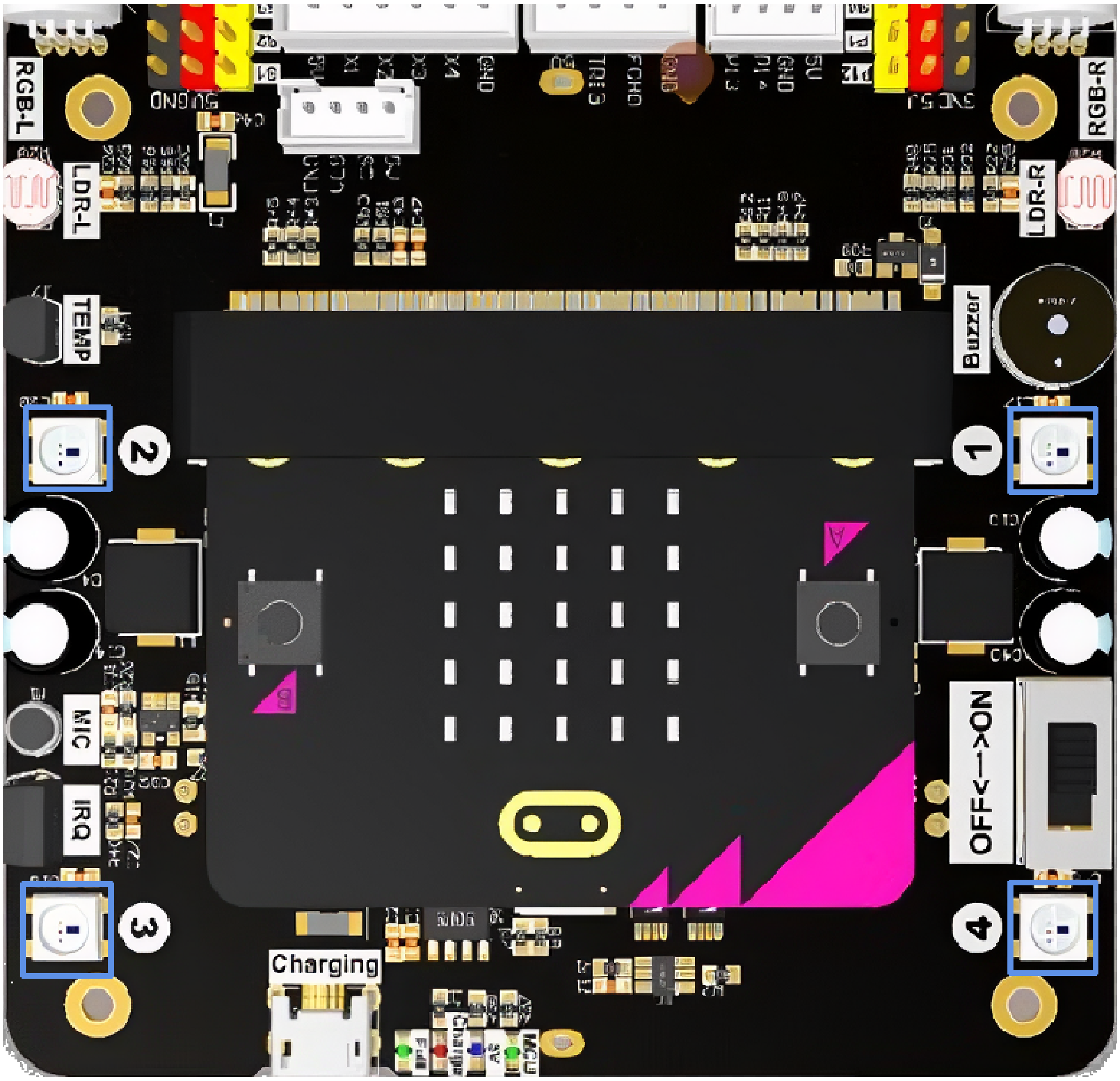
Colorful Breathing Light

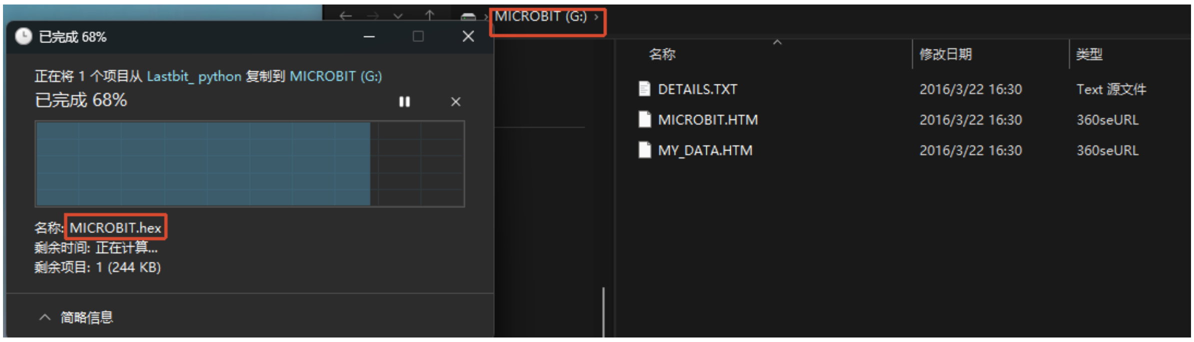
Feature Description

In the previous section, we introduced how to light up the colorful lights. In this section, we will learn how to use the breathing light effect based on the previous lesson.



Experimental Steps

Before flashing, make sure `LASTBIT.hex` has been flashed; otherwise, an error will occur and the module cannot be imported.



1. Before flashing, switch the switch to the OFF position and connect the computer to the microbit main board using a microUSB cable. **Note: it should be the Microbit interface, not the expansion board's Charging port.**

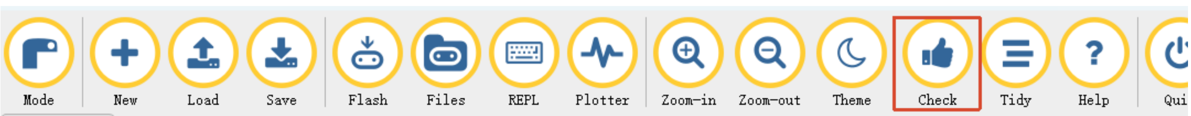
2. Open the Mu software. Here you can directly copy the program source code we provide; the operation steps are as follows. You can also enter the code yourself in the editor window. Note! All English characters and symbols should be entered in English input mode. Use the Tab key for indentation, and end the last line with a blank line.

   Click the `New` button in the toolbar at the top left, and a file titled untitled will appear.

3. Copy the content of the `Code Implementation` section provided below into the current file. You will notice a small red dot after the title bar, indicating that the code content has been modified but is not yet saved.

Whether or not you save does not affect the code check and flashing.

4. At this point, you can click `Check` to check whether the code format has any problems.



5. If you have connected the MICROBIT to the PC in advance following the steps above and have flashed `LASTBIT.hex`, the next step is to download the current program to the Microbit.

Click the `Flash` tool in the toolbar to start flashing!

During flashing, the orange-yellow indicator light near the Microbit main board will blink frequently, and the editor will show `Copied code onto micro:bit.` at the bottom, indicating that the flashing is complete. After flashing, the indicator light will no longer blink.

Code Implementation

```
# 4.Colorful_Breathing_Light
from microbit import *
from lastbit import LASTBIT
# import random to generate random numbers
import random

Lastbit = LASTBIT()

# Turn off all lights
Lastbit.all_lights_off()

while True:
    #  generate random RGB values
    r = random.randint(0, 255)
    g = random.randint(0, 255)
    b = random.randint(0, 255)
    #  breathing effect
    Lastbit.ws2812_breath(r, g, b, 3000)
```

`ws2812_breath(r, g, b, time)`: the first three parameters are the RGB values, and the last parameter `time` is used to adjust the duration of the breathing light effect, in milliseconds (ms).

Experimental Phenomenon

After the program is flashed, the colorful lights at the rear of the car will show a breathing light effect, completing one breath every three seconds. The color will change each time it lights up.